

Introduction

Large-scale convex optimization deals with problems whose dimension n is huge, yet we require iterative algorithms whose number of iterations is $\ll n$. Convexity is one of the very few structural properties that render such problems “solvable” both theoretically and practically.

Mathematical Optimization Problem

$$\begin{aligned} & \text{minimize} && f(x) \\ & \text{s.t.} && g_i(x) \leq 0, \quad i = 1, \dots, n_g, \\ & && h_i(x) = 0, \quad i = 1, \dots, n_h. \end{aligned} \quad (1)$$

- $x \in \mathbb{R}^n$ – decision variable (algorithms also work for $n \rightarrow \infty$); - f – objective; - $\mathcal{C} = \{\xi \in \mathbb{R}^n : g(\xi) \leq 0, h(\xi) = 0\}$ – feasible set.

Existence of a Minimizer

A minimizer exists when f is lower-semicontinuous, $f(x) \rightarrow +\infty$ as $|x| \rightarrow \infty$, and $\mathcal{C}$ is closed. In this case we write

$$\min_{x \in \mathcal{C}} f(x) \quad \text{and} \quad x^* \in \arg \min_{x \in \mathcal{C}} f(x).$$

First-Order Iterative Algorithms

All methods considered in the course are first-order:

$$x_{k+1} \leftarrow \text{function of } \{(f(x_j), \nabla f(x_j))\}_{j=0}^k.$$

The analytic complexity (number of oracle calls needed to reach accuracy ε) is the central quantity; arithmetic complexity (total floating-point operations) is secondary.

Curse of Dimensionality

For the class of L^∞ -Lipschitz problems on the unit hypercube, any algorithm may require

$$N \geq \left(\left\lfloor \frac{L}{2\varepsilon} \right\rfloor \right)^n - 1$$

oracle calls. Hence, without convexity or other structure, even modest accuracy is impossible for large n .

Convexity

Problem (1) is convex when f and all g_i are convex and all h_i are affine. In the convex case every local minimizer is global, which is the key property that makes large-scale problems tractable.

Convex sets and convex functions

Definition 1 (Convex Set). A set $\mathcal{C}$ is convex if and only if $\forall x, y \in \mathcal{C}$ and $\forall \theta \in [0, 1]$: $\theta x + (1 - \theta)y \in \mathcal{C}$.

Examples of convex sets:

- hyperplane $\{x \in \mathbb{R}^n \mid a^T x = b\}$
- half-space $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$
- polyhedron $\{x \in \mathbb{R}^n \mid Ax \preceq b, Cx = d\}$
 $A \in \mathbb{R}^{q \times n}, C \in \mathbb{R}^{r \times n}, b \in \mathbb{R}^q, d \in \mathbb{R}^r$
- ...more...

Operations that preserve convexity (sets)

- **Intersection** $\mathcal{C}_1, \mathcal{C}_2$ convex $\Rightarrow \mathcal{C}_1 \cap \mathcal{C}_2$ convex
- **Image under affine map** $\mathcal{C} \subseteq \mathbb{R}^n$ convex $\Rightarrow \{Ax + b \mid x \in \mathcal{C}\}$ convex
- inverse image of an affine map: ...

Separating Hyperplane Theorem

Theorem 1. $\mathcal{C} \subseteq \mathbb{R}^n$ non-empty closed convex set, $y \notin \mathcal{C} \Rightarrow \exists a \neq 0, b \in \mathbb{R}$ s. t. $a^T x + b < a^T y + b, \forall x \in \mathcal{C}$

Proof. Claim $\exists \hat{x} \in \mathcal{C}$ s.t. $|\hat{x} - y| \leq |x - y| \quad \forall x \in \mathcal{C}$

Proof of claim $|x - y|$ has bounded level sets, $\mathcal{C}$ is non-empty and closed $\Rightarrow \exists \hat{x} \in \arg \min_{x \in \mathcal{C}} |x - y|$

Hyperplane, we choose $a := y - \hat{x}, b := -a^T \hat{x} = -(y - \hat{x})^T \hat{x}$

As a result, $a^T x + b = (y - \hat{x})^T (x - \hat{x})$ and therefore $a^T y + b = |y - \hat{x}|^2 > 0$. The following claim shows that the hyperplane $a^T y + b$ separates $\mathcal{C}$ and y .

Claim $a^T y + b \leq 0 \quad \forall x \in \mathcal{C}$

Proof of claim Assume not. $\rightarrow \exists x \in \mathcal{C}$ s.t. $(y - \hat{x})^T (x - \hat{x}) > 0$

PARAMETRIZE θ

Contradiction $\hat{x}$ nearest point to y

(Details in Lecture notes) □

Corollary 1. A closed convex set $\mathcal{C} \subseteq \mathbb{R}^n$ is the intersection of the closed half-spaces that contain $\mathcal{C}$.

Proof. $\mathcal{S}$ intersection of closed half-spaces that contain $\mathcal{C}$

1) $\mathcal{C} \subseteq \mathcal{S} : x \in \mathcal{C} \Rightarrow x$ is contained in every half-spaces that contains $\mathcal{C} \Rightarrow x$ is also contained in the intersections of half-spaces that contains $\mathcal{C} \Rightarrow x \in \mathcal{S}$

2) $\mathcal{S} \subseteq \mathcal{C} : \text{Assume not} \rightarrow \exists \hat{x} \in \mathcal{S}$ with $\hat{x} \notin \mathcal{C}$. By the Separating Hyperplane Theorem there exists a hyperplane that separates $\hat{x}$ from $\mathcal{C}$. That means there exists a closed half-space that contains $\mathcal{C}$ but not $\hat{x}$, hence $\hat{x} \notin \mathcal{S}$, contradiction. □

Support function

Idea represent any closed convex set by its supporting hyperplanes

Support Function: $\sigma_C(a) = \sup_{x \in C} a^T x$

CALCULATION EXAMPLE

If we know the $\sigma_C(a)$, we arrive at at

$$C = \bigcap_{a \in \mathbb{R}^n} \{x \in \mathbb{R}^n \mid a^T x - \sigma_C(a) \leq 0\}$$

$$= \{x \in \mathbb{R}^n \mid \sup_{a \in \mathbb{R}^n} a^T x - \sigma_C(a) \leq 0\}$$

Definition 2. A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex if and only if its epigraph is a convex set, where

$$\text{epi}(f) := \{(x, t) \in \mathbb{R}^{n+1} \mid f(x) \leq t\}$$

→ this provides a link between convex sets and functions

Operations that preserve convexity (functions)

- the pointwise maximum of convex functions is convex
- the sum of convex functions is convex
- $f(Ax + b)$ is convex if f is convex

How to check if f is convex?

- if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ twice differentiable, $\partial^2 f / \partial x^2 \succeq 0 \forall x \in \mathbb{R}^n$
- if $g : \mathbb{R} \rightarrow \mathbb{R}$ with $g(t) = f(x + tv)$ convex in $t \forall x, v \in \mathbb{R}^n$, then f is convex
- composition of simple convex function with convexity preserving operations

Extended real numbers $\bar{\mathbb{R}} = \mathbb{R} \cup \{+\infty, -\infty\}$

Indicator function $\psi_C(x) := \begin{cases} +\infty & \text{if } x \notin C \\ 0 & \text{if } x \in C \end{cases}$

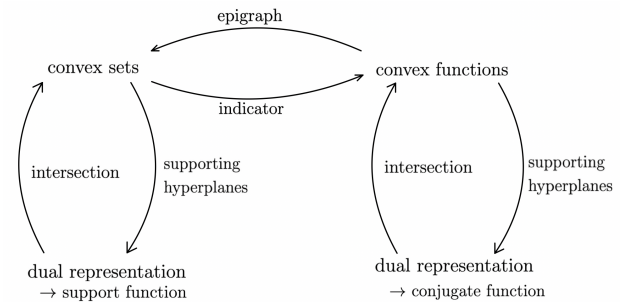
→ this provides another link between convex sets and functions

We can write $\min_{x \in C} f(x)$ as $\min_{x \in \mathbb{R}^n} f(x) + \psi_C(x)$

Definition 3 (3). $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is called proper if f is bounded below and if $\exists x \in \mathbb{R}^n$ s. t. $f(x) < \infty$

Definition 4 (Legendre Transformation). The conjugate function of $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is defined as $f^*(y) = \sup_{x \in \mathbb{R}^n} y^T x - f(x)$

Summary of Concepts



Convex Sets and Convex Functions

Definition 5 (Convex set). A set C is convex if and only if

$$\theta x + (1 - \theta)y \in C \quad \forall x, y \in C, \theta \in [0, 1].$$

Examples. Hyperplanes, half-spaces, polyhedra, convex hulls, positive-semidefinite cone $\mathbb{S}_+^n$.

Operations Preserving Convexity

Intersection, affine images/pre-images, Minkowski sums, Cartesian products, conic combinations.

Separating Hyperplane Theorem

Let $C \subset \mathbb{R}^n$ be nonempty, closed and convex and $y \notin C$. Then there exist $a \neq 0$ and $b \in \mathbb{R}$ such that

$$a^T x + b \leq a^T y + b \quad \forall x \in C.$$

(The proof proceeds by projecting y onto C and constructing the normal $a = y - \hat{x}$.)

Corollary. Every closed convex set is the intersection of the closed half-spaces that contain it.

Support Function

The support function

$$\sigma_C(a) = \sup_{x \in C} a^T x$$

completely characterises a closed convex set:

$$C = \bigcap_{a \in \mathbb{R}^n} \{x \mid a^T x \leq \sigma_C(a)\}.$$

Convex Functions

A function $f : \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is convex if and only if its epigraph is a convex set. Equivalently,

$$f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y).$$

Operations preserving convexity. Nonnegative linear combinations, pointwise maxima, composition with affine maps, perspective, etc.

Indicator function. $\psi_C(x) = 0$ if $x \in C$ and $+\infty$ otherwise. Then

$$\min_{x \in C} f(x) = \min_{x \in \mathbb{R}^n} f(x) + \psi_C(x).$$

Conjugate. $f^*(y) = \sup_x (y^\top x - f(x))$. For closed convex proper f we have $y \in \partial f(x) \iff x \in \partial f^*(y)$.

Convex Sets and Convex Functions

Definition 6 (Convex set). A set $C \subseteq \mathbb{R}^n$ is convex if for all $x, y \in C$ and $\theta \in [0, 1]$ we have $\theta x + (1 - \theta)y \in C$.

Examples: hyperplanes $\{x \mid a^\top x = b\}$, half-spaces $\{x \mid a^\top x \leq b\}$, polyhedra $\{x \mid Ax \preceq b, Cx = d\}$, convex hulls, and balls.

Operations preserving convexity (sets): intersection, affine images and pre-images, Minkowski sums, scaling, Cartesian products.

Separating and Supporting Hyperplanes

Theorem 2 (Separating Hyperplane). Let $C \subseteq \mathbb{R}^n$ be nonempty, closed, and convex, $y \notin C$. Then there exist $a \neq 0, b \in \mathbb{R}$ such that $a^\top x + b < a^\top y + b$ for all $x \in C$.

Key ideas (Lecture 2). Let $\hat{x} = \arg \min_{x \in C} \|x - y\|$ (exists by closedness and coercivity of the distance). Set $a = y - \hat{x}, b = -a^\top \hat{x}$. The hyperplane $a^\top x + b = 0$ separates because any violation would allow a point on the line segment closer to y than $\hat{x}$, a contradiction. $\square$

A closed convex set is the intersection of all closed half-spaces containing it (Corollary from separating theorem). This yields a dual representation via the *support function*

$$\sigma_C(a) = \sup_{x \in C} a^\top x.$$

Then

$$C = \{x \mid \sup_a (a^\top x - \sigma_C(a)) \leq 0\}.$$

Convex Functions

A function $f: \mathbb{R}^n \rightarrow \bar{\mathbb{R}}$ is convex if and only if its epigraph

$$\text{epi}(f) = \{(x, t) \mid f(x) \leq t\}$$

is a convex set. This links sets and functions.

Operations preserving convexity (functions): pointwise supremum, nonnegative weighted sums, composition with affine maps $f(Ax + b)$, infimal convolution.

Verification of convexity: - Hessian $\nabla^2 f(x) \succeq 0$ for all x (twice differentiable case). - Restriction to every line $t \mapsto f(x + tv)$ is convex. - Composition of known convex functions via the rules above.

The *indicator function* of a convex set C is

$$\psi_C(x) = \begin{cases} 0 & x \in C, \\ +\infty & \text{otherwise.} \end{cases}$$

Then $\min_{x \in C} f(x) = \min_x (f(x) + \psi_C(x))$.

A function is *proper* if it is bounded below and finite somewhere.

The *conjugate* (Legendre transform) is

$$f^*(y) = \sup_x (y^\top x - f(x)).$$

It represents the epigraph by supporting hyperplanes with slope $(y, -1)$. The biconjugate theorem (Lecture 2) states that if f is proper, convex, and $\text{epi}(f)$ closed, then $f^{**} = f$ (and always $f \geq f^{**}$).

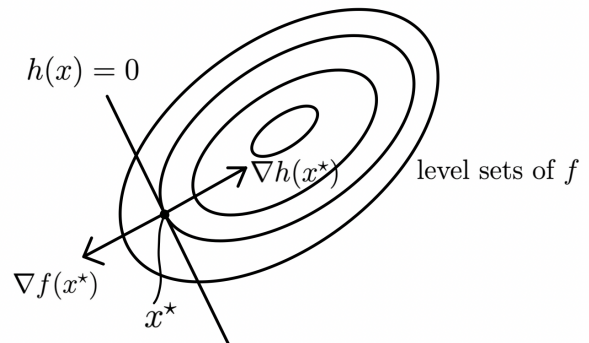
A useful summary diagram (Lecture 2, page 8) connects convex sets $\leftrightarrow$ supporting hyperplanes $\leftrightarrow$ support functions, and convex functions $\leftrightarrow$ epigraph $\leftrightarrow$ conjugate functions.

Recitation examples (log-sum-exp as smooth approximation of the max, log-barrier for inequalities, norms, nonnegative weighted sums) illustrate these concepts in practice.

KKT Conditions and Lagrange Duality

Example

Optimization problem: $\min_{x \in \mathbb{R}^2} f(x)$ s.t. $h(x) = 0$



We note the following: $\nabla f(x^*)$ and $\nabla h(x^*)$ are colinear

$$\Leftrightarrow \exists \nu^* \in \mathbb{R} : \nabla f(x^*) + \nu^* \nabla h(x^*) = 0$$

$\Leftrightarrow f(x) + \nu^* h(x)$ is stationary at x^* , where ν^* can be interpreted as cost of violating constraint

Generalization

Consider

$$f^* = \inf_x f(x) \quad \text{s.t.} \quad h(x) = 0, g(x) \leq 0.$$

The Lagrangian is

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \lambda^\top g(x) + \nu^\top h(x), \quad \lambda \geq 0.$$

Weak Duality

$d(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu) \leq f^*$ for all feasible multipliers.

Strong Duality

If the problem is convex and Slater's condition holds ($\exists \hat{x}$ with $g(\hat{x}) < 0, h(\hat{x}) = 0$), then there exist $\lambda^* \geq 0, \nu^*$ such that $d(\lambda^*, \nu^*) = f^*$.

KKT Optimality Conditions

Under convexity + Slater, x^* is primal optimal and (λ^*, ν^*) is dual optimal if and only if

$$\begin{aligned} \nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0 && \text{(stationarity),} \\ g(x^*) &\leq 0, \quad h(x^*) = 0 && \text{(primal feasibility),} \\ \lambda^* &\geq 0 && \text{(dual feasibility),} \\ \lambda^{*\top} g(x^*) &= 0 && \text{(complementary slackness).} \end{aligned}$$

The last equality together with strong duality yields

$$\sup_{\lambda \geq 0, \nu} d(\lambda, \nu) = \inf_{x \in C} f(x).$$

Subdifferential (Nonsmooth Case)

For a convex function the subdifferential at $\bar{x}$ is

$$\partial f(\bar{x}) = \{v \mid f(x) \geq f(\bar{x}) + v^\top (x - \bar{x}) \quad \forall x\}.$$

A point x^* minimises f if and only if $0 \in \partial f(x^*)$. When f is differentiable we recover the classical gradient condition.

Convex Optimization Problem Classes

Feasibility Problems

Reduce to checking whether the optimal value of an auxiliary problem with slack s is non-positive.

Linear Programming (LP)

$$\min c^\top x \quad \text{s.t.} \quad Ax - b \geq 0, \quad x \geq 0.$$

Its dual is again an LP. Optimal solutions, when they exist, lie at vertices of the polyhedron.

Quadratic Programming (QP)

$$\min \frac{1}{2} x^\top P x + q^\top x \quad \text{s.t.} \quad Gx \leq h, \quad Ax = b.$$

Convex when $P \succeq 0$.

Second-Order Cone Programming (SOCP)

$$\min f^\top x \quad \text{s.t.} \quad \|A_i x + b_i\| \leq c_i^\top x + d_i.$$

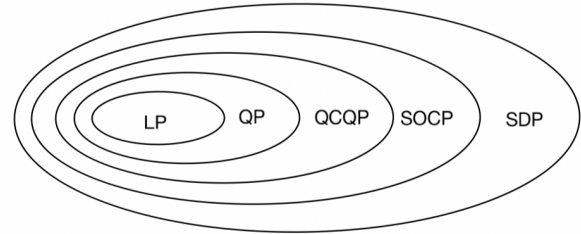
The second-order cone $C_{n+1} = \{(x, t) : \|x\| \leq t\}$ is self-dual and admits efficient projection.

Semidefinite Programming (SDP)

$$\min \langle C, X \rangle \quad \text{s.t.} \quad \langle F_i, X \rangle = b_i, \quad X \succeq 0.$$

SDP is the most general convex problem class that can still be solved in polynomial time (to any fixed accuracy) by interior-point methods.

Hierarchy of problem classes. $LP \subset QP \subset SOCP \subset SDP$.



Gradient Methods – Part I

Definition 7 (Smoothness). f is L -smooth if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$. Consequently,

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|x - y\|^2.$$

Definition 8 (Strong convexity). f is μ -strongly convex if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2.$$

Gradient Descent on Quadratics

For $f(x) = \frac{1}{2} x^\top H x + b^\top x$ with $H \succeq \mu I$ and $H \preceq L I$ the iteration becomes

$$x_{k+1} - x^* = (I - TH)(x_k - x^*).$$

The optimal step size $T^* = 2/(L + \mu)$ yields the contraction factor

$$\rho = \frac{L - \mu}{L + \mu} = \frac{\kappa - 1}{\kappa + 1}, \quad \kappa = L/\mu.$$

Hence $N \geq (\kappa + 1)/2 \cdot \log(\|x_0 - x^*\|/\varepsilon)$ iterations suffice.

Momentum Methods (Nesterov / Heavy-Ball)

The two-step recurrence

$$q_{k+1} = q_k + T p_{k+1}, \quad p_{k+1} = (1 - 2dT) p_k - T \nabla f(q_k + \beta p_k) / L$$

with the classical parameter choices realises the optimal rate

$$\left(1 - \frac{1}{\sqrt{\kappa}}\right)^N$$

(up to a constant factor). This rate is optimal among all first-order methods on the class of smooth strongly-convex functions.

Gradient Methods - Part II

Projection and Projected Gradient Descent

For a closed convex set $\mathcal{C}$ the projection is

$$\text{prox}_{\mathcal{C}}(x) = \arg \min_{y \in \mathcal{C}} \frac{1}{2} \|x - y\|^2.$$

It is firmly non-expansive:

Lemma 1. $|\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y)|^2 \leq (\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y))^T (x - y)$, implying $\|\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y)\| \leq \|x - y\|$.

The projected gradient step is $x_{k+1} = \text{prox}_{\mathcal{C}}(x_k - T \nabla f(x_k))$. For L -smooth μ -strongly convex f and $T = 2/(L + \mu)$ the same linear rate as the unconstrained case holds:

$$\|x_N - x^*\| \leq \left(1 - \frac{2}{L + \mu}\right)^N \|x_0 - x^*\|.$$

Proof (Lecture 6): the fixed point property $x^* = \text{prox}_{\mathcal{C}}(x^* - T \nabla f(x^*))$ follows from the optimality condition $T \nabla f(x^*) + \partial \psi_{\mathcal{C}}(x^*) \ni 0$; the contraction follows by non-expansiveness of the prox and the same contraction of $I - TH$ as in the unconstrained quadratic case.

Projections are cheap for Euclidean balls ($\text{prox}(y) = y / \max(1, \|y\|)$), ℓ_1 -balls (soft-thresholding after sorting), probability simplices, and low-dimensional affine subspaces.

Smooth but Not Strongly Convex ($\mu = 0$)

Add a small strongly convex regularizer:

$$\tilde{f}(x) = f(x) + \frac{\mu}{2} \|x - x_0\|^2.$$

Lemma (Lecture 6): $\tilde{f}$ is $(L + \mu)$ -smooth and μ -strongly convex, $f(x) - f(x^*) \leq \tilde{f}(x) - \tilde{f}(x^*) + (\mu/2) \|x^* - x_0\|^2$, and $\|\tilde{x}^* - x_0\| \leq \|x^* - x_0\|$.

Applying Nesterov's accelerated method to $\tilde{f}$ and optimizing the regularization parameter $\mu \approx (2L \ln N)/N$ yields (after using $e^\epsilon \geq 1 + \epsilon$)

$$f(x_N) - f(x^*) \leq \left(\frac{2L}{N^2} + \frac{9L \ln(N)^2}{2N^2}\right) \|x^* - x_0\|^2.$$

A refined analysis gives the optimal accelerated rate (Beck–Teboulle FISTA):

Theorem 3. The scheme $q_{k+1} = q_k + p_{k+1}$, $p_{k+1} = \beta_k p_k - g_f(q_k + \beta_k p_k)/L$ with $\beta_k = t_k/t_{k+1}$, $t_{k+1} = (1 + \sqrt{1 + 4t_k^2})/2$, $t_0 = 1$ satisfies

$$f(q_N) - f(x^*) \leq \frac{2L \|q_0 - x^*\|^2}{(N+1)^2}.$$

A matching lower bound of order $1/N^2$ exists.

Thus, roughly $N \sim \sqrt{L \|x^* - x_0\|^2 / \epsilon}$ iterations suffice for ϵ -accuracy.

Non-Smooth Convex Objectives

If f is convex and L_f -Lipschitz (but not differentiable), constant-step gradient descent can oscillate (example $f(x) = |x|$). Use a diminishing stepsize. The projected subgradient method with $T = RL_f/\sqrt{N}$ on a set of radius R yields

$$f\left(\frac{1}{N} \sum_{k=0}^{N-1} x_k\right) - f(x^*) \leq \frac{RL_f}{\sqrt{N}}.$$

Proof (Lecture 6): combine the subgradient inequality $f(x^*) \geq f(x_k) + g_k^T (x^* - x_k)$, the non-expansiveness of the prox (distance to x^* decreases), and telescope; convexity of f lets us bound the average.

The plot of error vs. iteration (Lecture 6) clearly shows linear, $1/k$, $1/k^2$, and $1/\sqrt{k}$ regimes.

Recitation remarks on adaptive stepsize rules, normalized gradient descent, and practical tuning close the algorithmic part. These rates, together with the theoretical foundations of the first four chapters, provide a complete toolkit for recognizing, formulating, and solving large-scale convex problems with first-order methods.

Gradient Methods – Part II

Projected Gradient Descent

Let $\text{prox}_{\mathcal{C}}(x) = \arg \min_{y \in \mathcal{C}} \frac{1}{2} \|x - y\|^2$. For a closed convex set the proximal operator is firmly non-expansive:

$$\|\text{prox}_{\mathcal{C}}(x) - \text{prox}_{\mathcal{C}}(y)\| \leq \|x - y\|.$$

The iteration $x_{k+1} = \text{prox}_{\mathcal{C}}(x_k - T \nabla f(x_k))$ with $T = 2/(L + \mu)$ converges exactly as the unconstrained method when f is smooth and strongly convex.

Cheap projections. Euclidean ball, ℓ_1 -ball, probability simplex, affine subspaces.

Non-Strongly-Convex Case

Regularise:

$$\tilde{f}(x) = f(x) + \frac{\mu}{2} \|x - x_0\|^2.$$

Apply accelerated gradient descent to $\tilde{f}$ and optimise the regularisation parameter $\mu \sim L \log N/N$. The resulting sub-linear rate is

$$f(x_N) - f(x^*) \leq O\left(\frac{L \|x_0 - x^*\|^2}{N}\right).$$

Hence $N \sim L \|x_0 - x^*\|^2 / \epsilon$ iterations are necessary and sufficient.

Non-Smooth Case

When f is merely Lipschitz (subgradient method) the best achievable rate on a bounded domain is $O(1/\sqrt{N})$. This rate is optimal.